

James Duxbury AfCIIS

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EDUCATION

Durham University

Expected Graduation: July 2026

Integrated Master of Engineering in Computer Science

Oct 2022 – July 2026

- On track for **First Class Honours**; strong performance in Software Engineering, Data Science and AI modules.
- Project Lead on IBM-sponsored software engineering project (grade 80%), implementing NPC and player logic in an agile, team-based environment.
- Achieved 98% in both full-stack **web development** courseworks.

SKILLS

Languages & Frameworks: Python, JavaScript/TypeScript, C/C++, C#, Haskell, React, Node.js, Jest

AI & Data Science: PyTorch, Scikit-Learn, OpenCV, Pandas, NumPy, Matplotlib

Cloud & Infrastructure: Git, GitHub, GitLab CI/CD, Docker, AWS, Vercel, Render, Neon, Linux

Development Practices: Agile, TDD, Pair/Mob Programming, Code Review

AI Development Tools: Windsurf, Cursor, Claude Code

PROFESSIONAL EXPERIENCE

Software Engineer

July 2025 – September 2025

Compare the Market - Internship

Peterborough

- Designed and developed a Vite + React dashboard enabling non-technical users to visualise and explore complex question sets across **9 products**, reducing reliance on developers for data retrieval and reducing analysis time.
- Engineered a recursive parsing algorithm to translate **JSONLogic** rules into natural-language bullet points, generalising across diverse logical expressions and business operators, facilitating improved knowledge transfer between technical and business teams.
- Led a mob-programming intern project to re-engineer a legacy internal tool, coordinating **collaborative development** while implementing modern UI updates, advanced filtering, and maintainability improvements for future system migration.
- Implemented and optimised **GitLab CI/CD pipelines**, enforcing automated testing and code quality checks, which improved maintainability and accelerated development workflows.
- Enhanced productivity by applying AI-assisted development tools (**Windsurf/Cursor**) to streamline debugging, refactor code for scalability, and auto-generate comprehensive test cases, enabling greater focus on steering system architecture and long-term maintainability.

Software Engineer - AI Trainer

Mar 2024 – Present

Data Annotation - Contract

Remote

- Debugged and optimised LLM-generated **Python, JavaScript, and SQL** code, improving accuracy and reliability of model outputs.
- Delivered structured feedback on syntax, logic, and explanatory text, directly enhancing **training data quality** for large-scale AI model fine-tuning.
- Applied systematic debugging and problem decomposition skills to resolve complex logical errors in AI-generated code.

Information Technology Support Technician

Oct 2021 – Present

Snap-on Tools

Remote

- Built and deployed a **Python automation tool** to audit and clean enterprise asset databases, cutting data errors and improving integrity across **500+ users**.
- Developed a Python GUI application to automate **Exchange 365 migrations** for **400+ users**, reducing manual effort by **70%** and ensuring smoother adoption.
- Supported development of a custom web-based **asset tracking system**, streamlining internal infrastructure and improving transparency for distributed teams.

PROJECTS

JSGrades: Qualification Tracker (In Progress) | *React, NodeJS, PostgreSQL* May 2025 – Present

- Developing a full-stack React + Node.js application for **degree grade forecasting, task tracking, and performance visualisation**.
- Implemented CI/CD with Vercel (frontend), Render (backend), and Neon (PostgreSQL), demonstrating cloud-native deployment pipelines.
- Implementing authentication with **Firebase**, testing with **Jest** and utilising **Qodana, ESLint and Prettier** to improve code quality, security and maintainability.

X-Ray Image Repair | *Python, OpenCV* May 2018 – May 2020

- Applied image processing and Computer Vision techniques to recover corrupted radiology scans, boosting a CNN classifier's accuracy from 55% to 90%.

The Disappearance | *Godot* Oct 2023 – May 2024

- Collaborated in an agile team to deliver an RPG prototype, with personal contributions to **character control logic and NPC interactions**.
- Demonstrated ability to apply software engineering principles in a game development context, balancing usability and performance.

CERTIFICATIONS AND MEMBERSHIPS

Microsoft Certified: Azure AI Fundamentals (2024)

ITIL 4 Foundation (2021)

IBM: Enterprise Design Thinking Practitioner; Data Science in Practice; Cloud: Envisioning Your Solution (2023)

Affiliate Member, Chartered Institute of Information Security (AfCIIS) (2020)